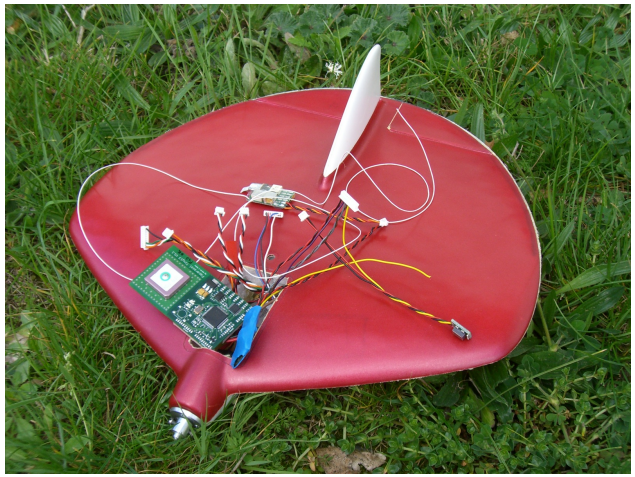
Yet Another Paparazzi

P. Brisset, M. Bronz, M. Gorraz and J. Tyler

ENAC

March 2008



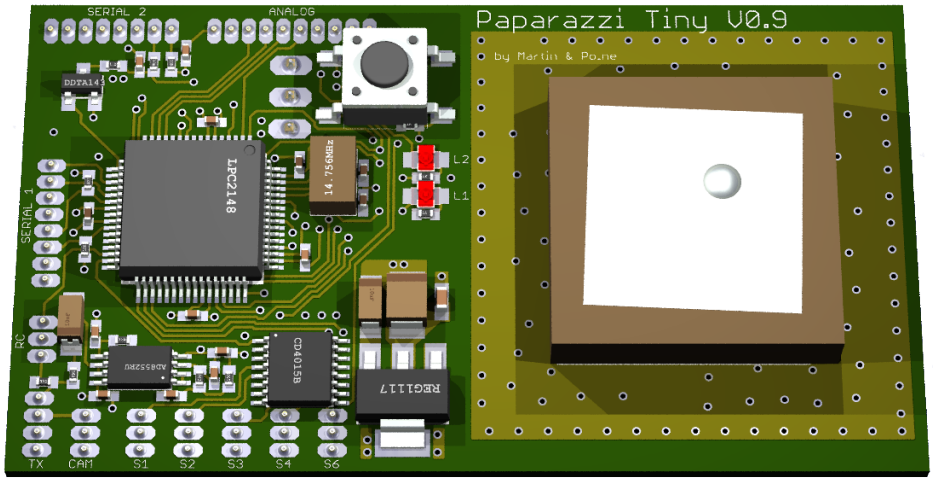


The Paparazzi System



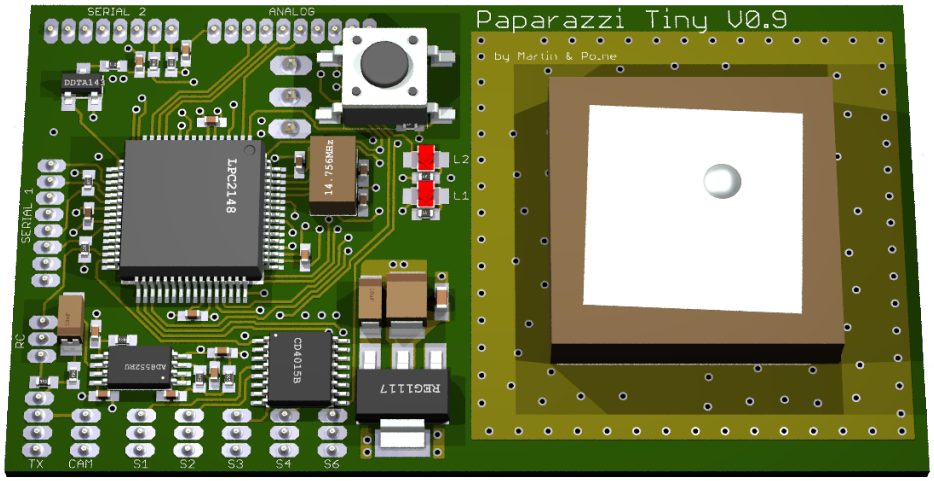
Autopilot

- Processor
- GPS
- Power supply
- 20g

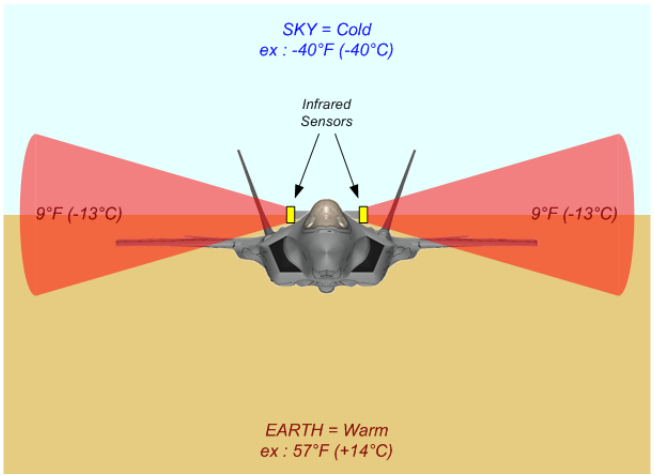


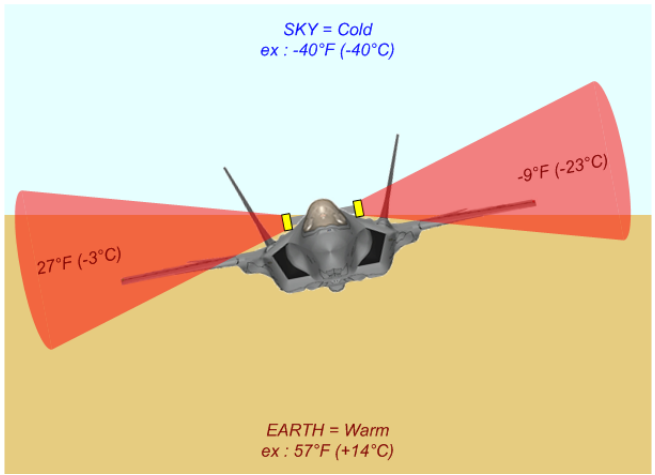
Autopilot

- Processor
- GPS
- Power supply
- 20g

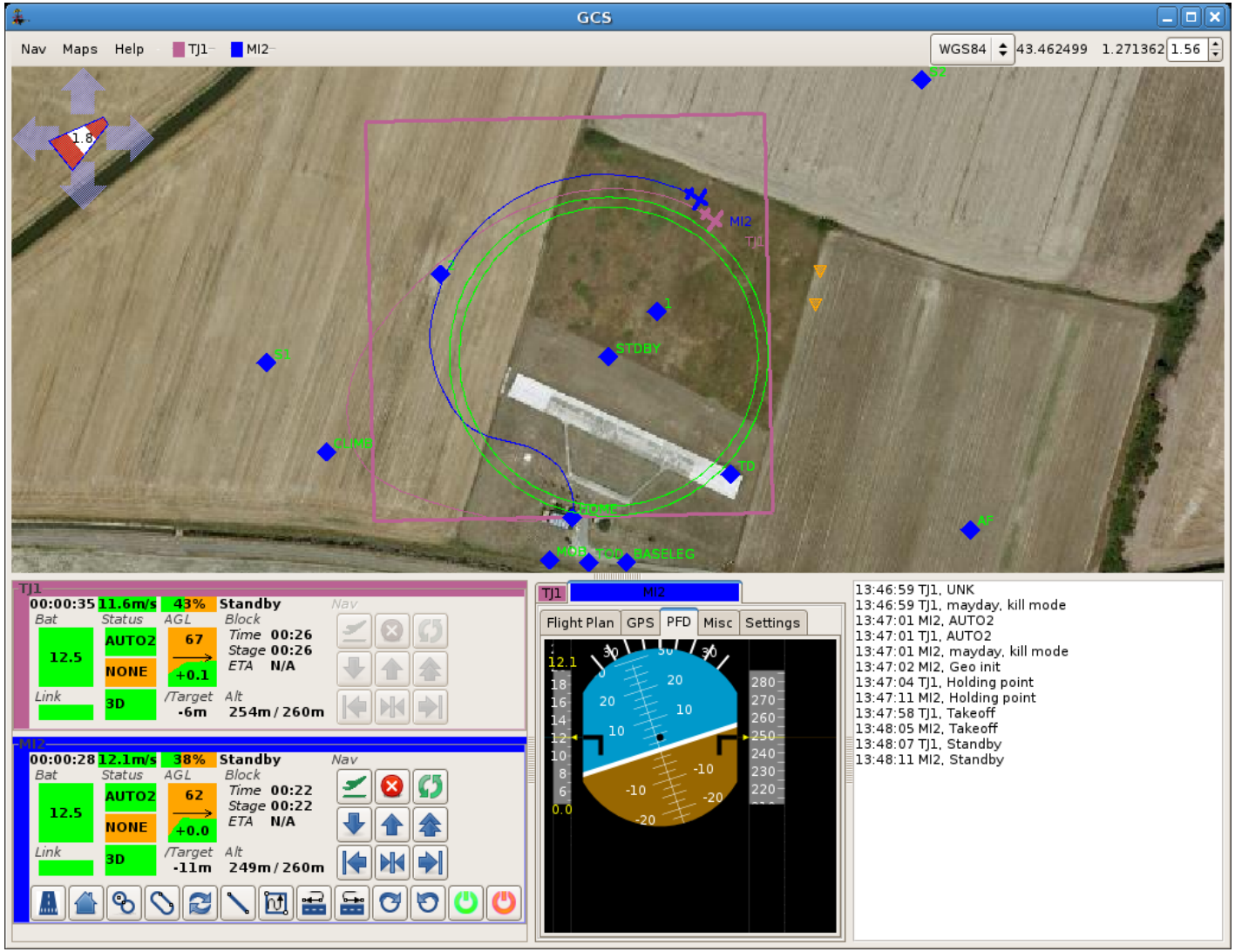


Attitude Estimation with Infrared

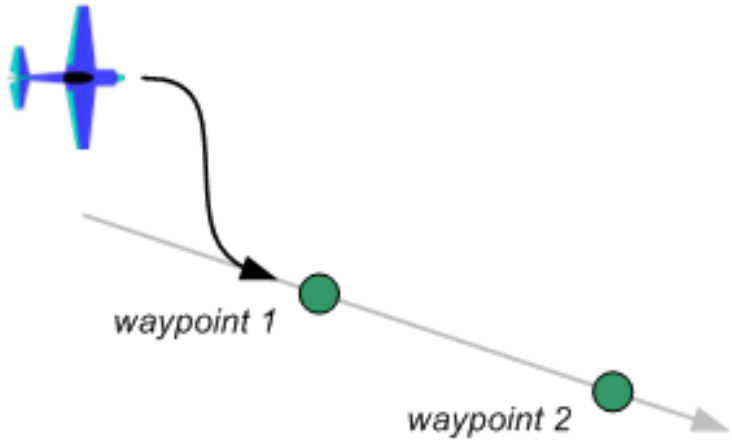




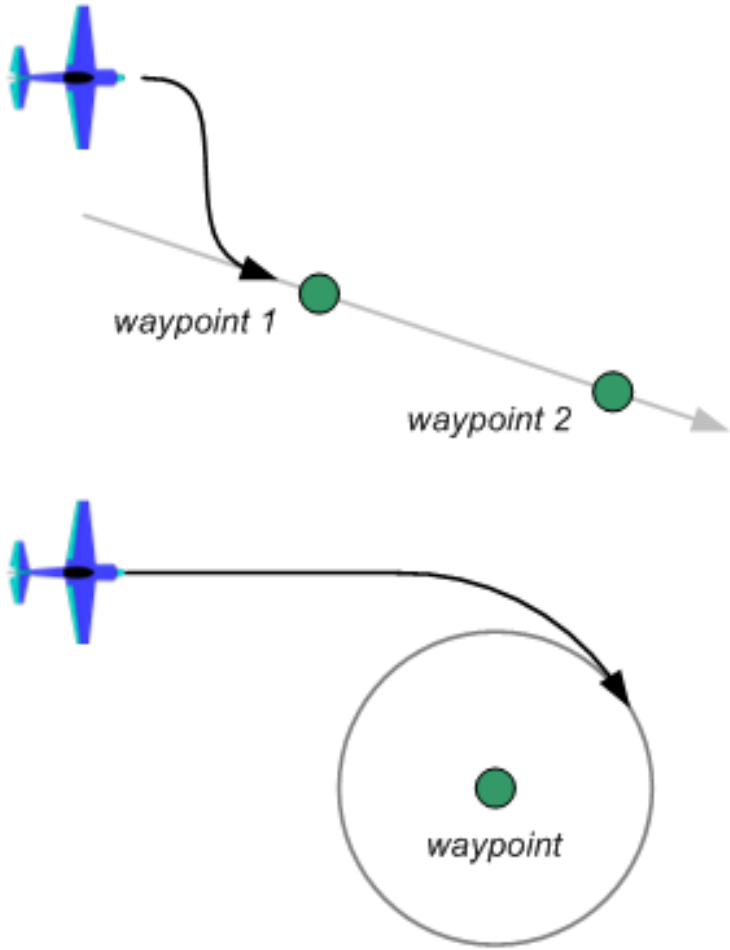
The GCS


GCS
Nav Maps Help TJ1 MI2
WGS84 43.462499 1.271362 1.56
MI2
TJ1
STDBY
CLIMB
TD
BASELEG
TJ1
00:00:35 11.6m/s 43% Standby
Bat Status AGL Block
12.5 AUTO2 67 Time 00:26
NONE +0.1 Stage 00:26
Link 3D /Target -6m Alt 254m/260m
ETA N/A
MI2
00:00:28 12.1m/s 38% Standby
Bat Status AGL Block
12.5 AUTO2 62 Time 00:22
NONE +0.0 Stage 00:22
Link 3D /Target -11m Alt 249m/260m
ETA N/A
Flight Plan GPS PFD Misc Settings
13:46:59 TJ1, UNK
13:46:59 TJ1, mayday, kill mode
13:47:01 MI2, AUTO2
13:47:01 TJ1, AUTO2
13:47:01 MI2, mayday, kill mode
13:47:02 MI2, Geo init
13:47:04 TJ1, Holding point
13:47:11 MI2, Holding point
13:47:58 TJ1, Takeoff
13:48:05 MI2, Takeoff
13:48:07 TJ1, Standby
13:48:11 MI2, Standby

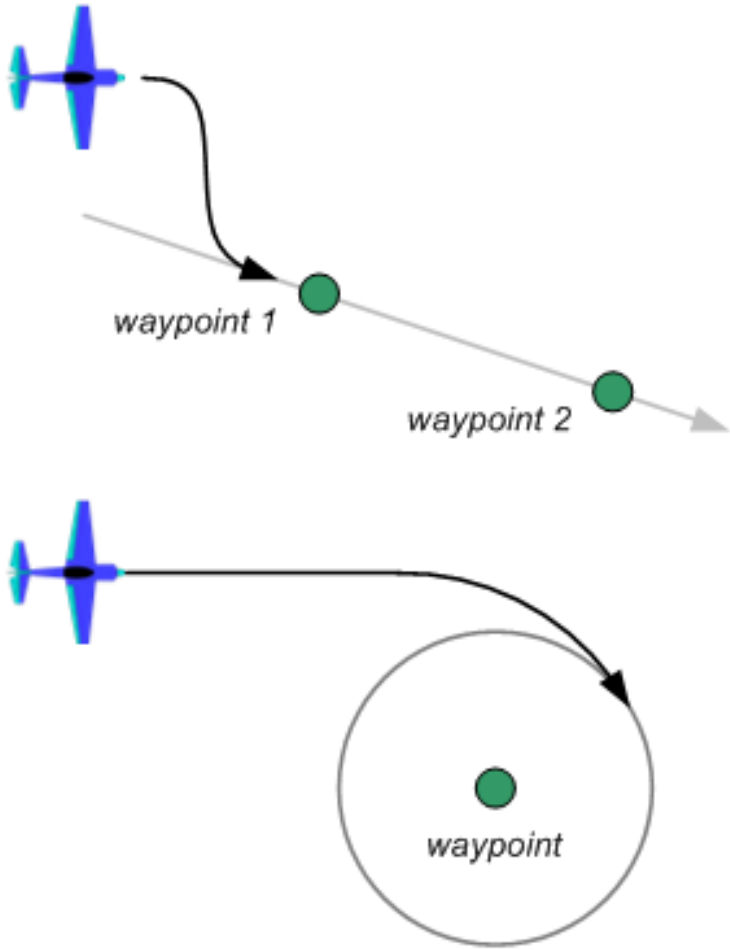
Flight Plan

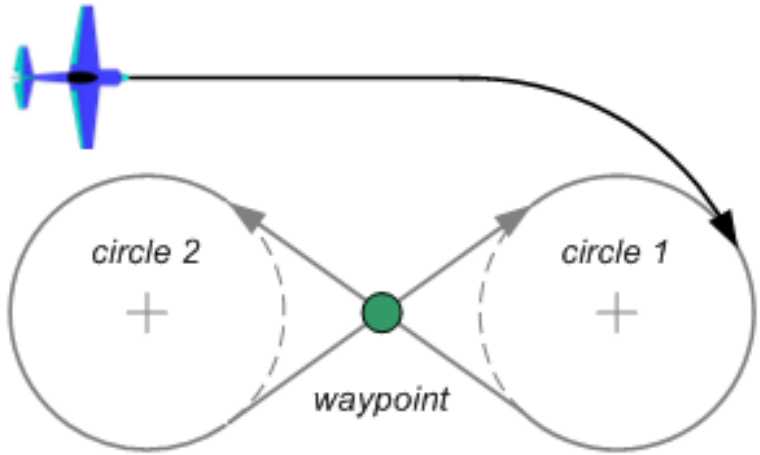


Flight Plan

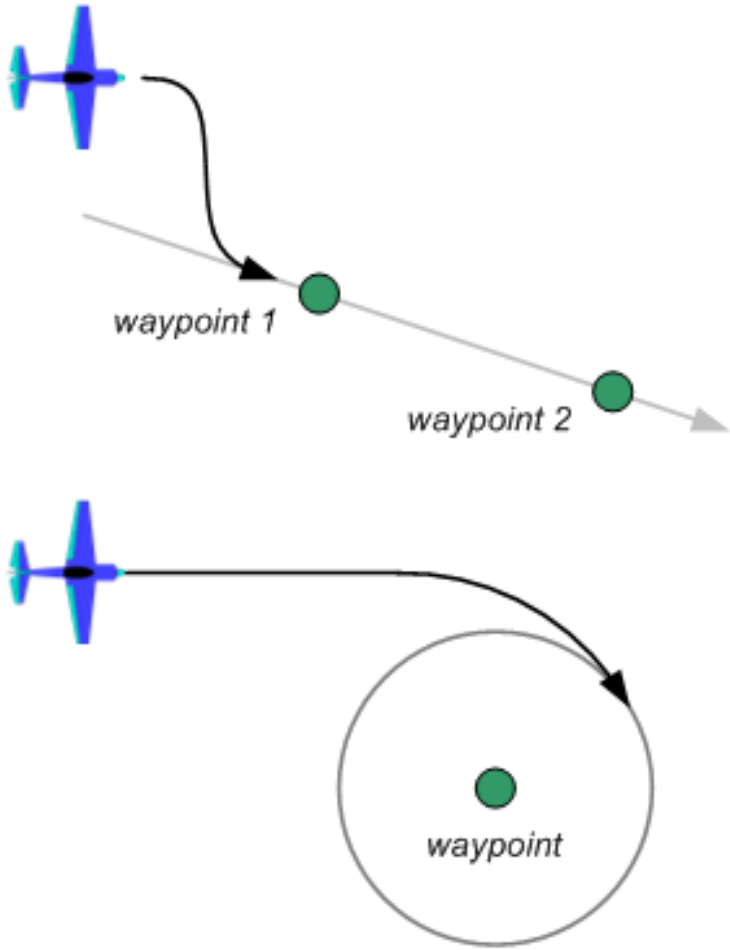


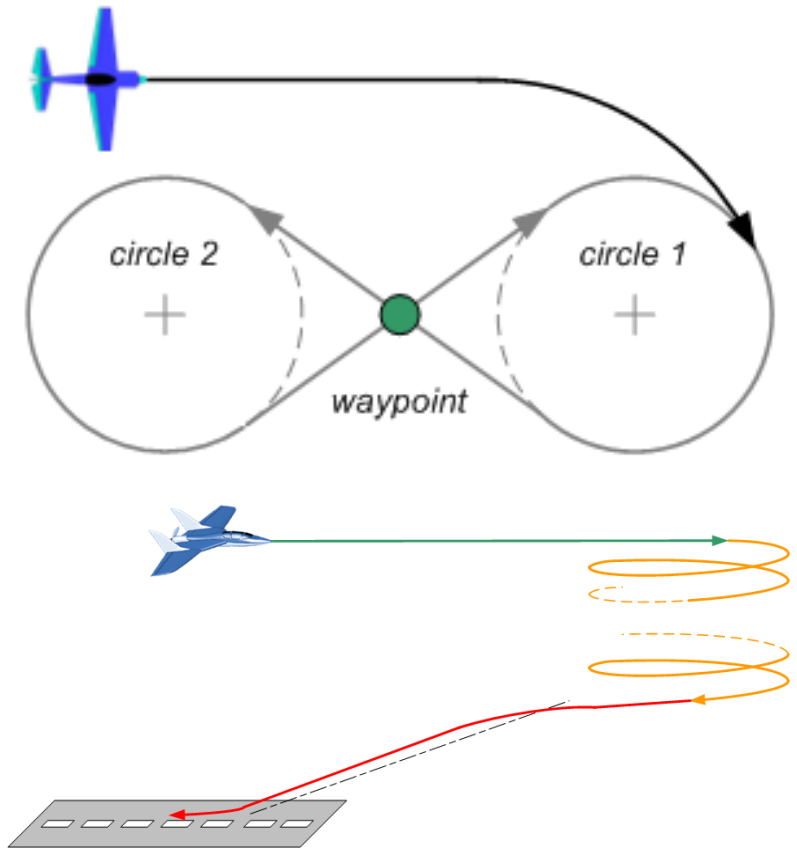
Flight Plan



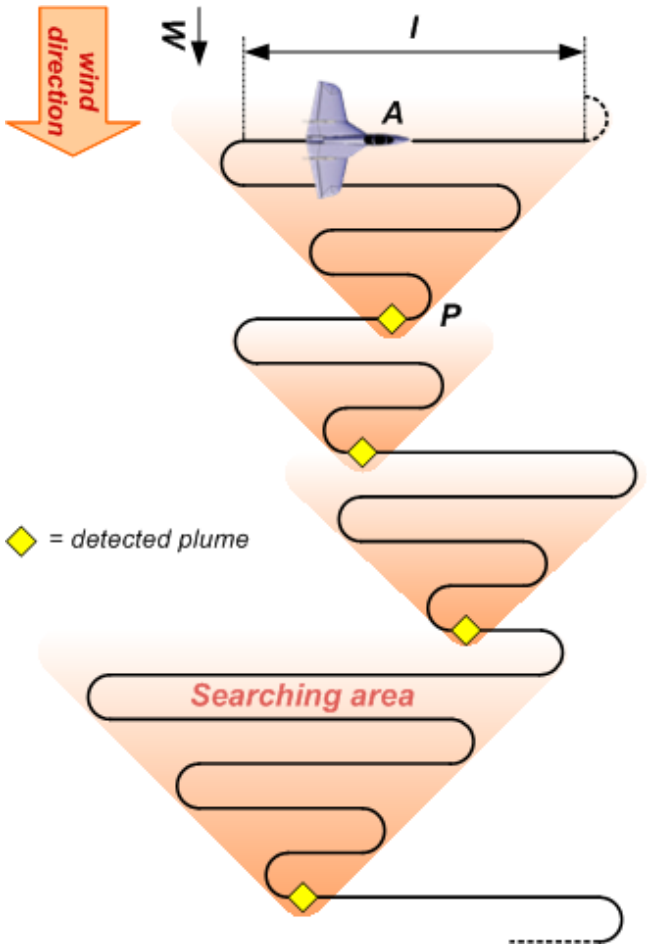


Flight Plan

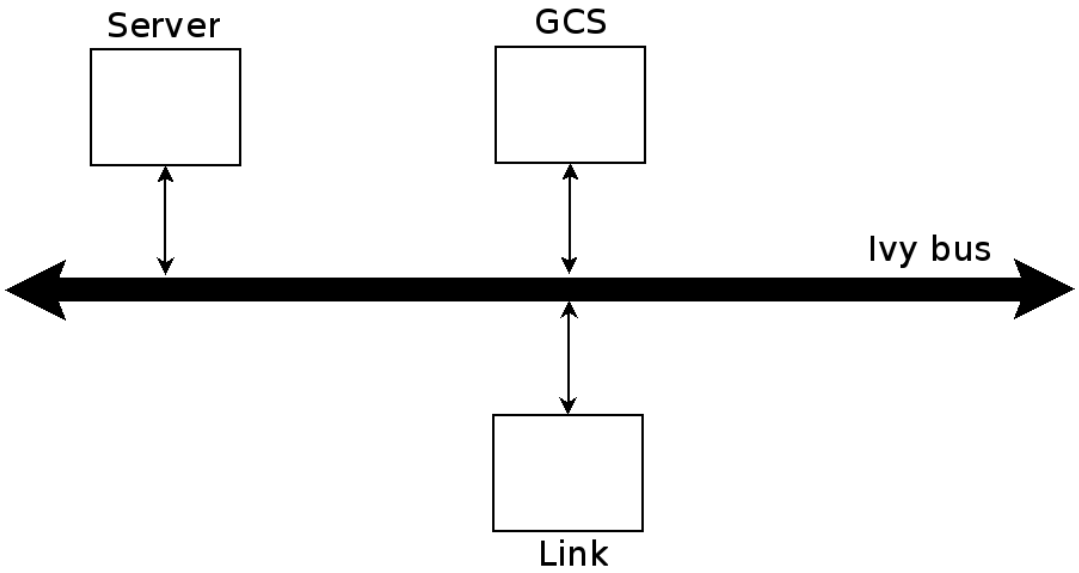




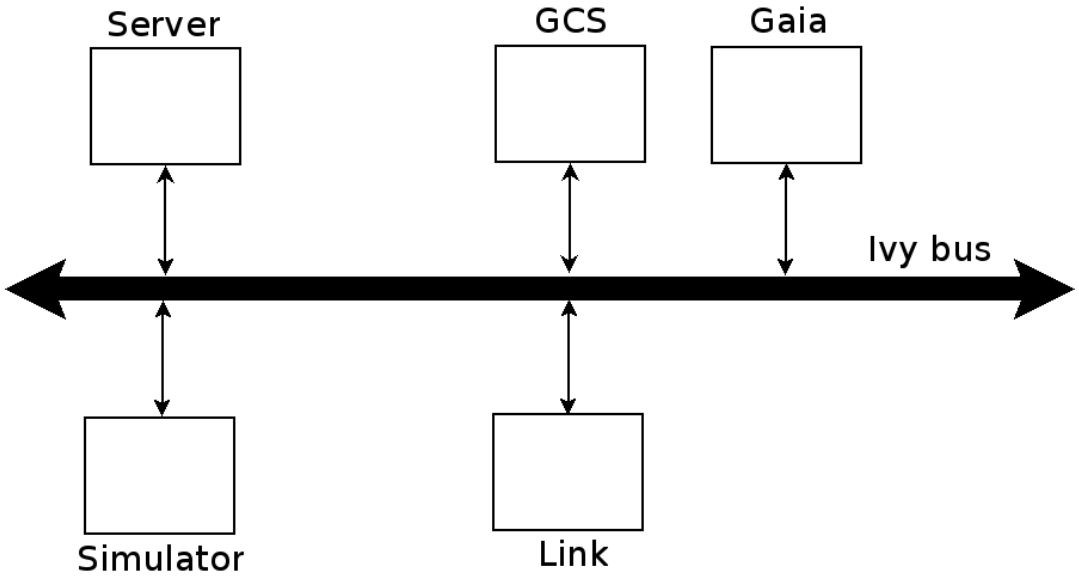
Flight Plan



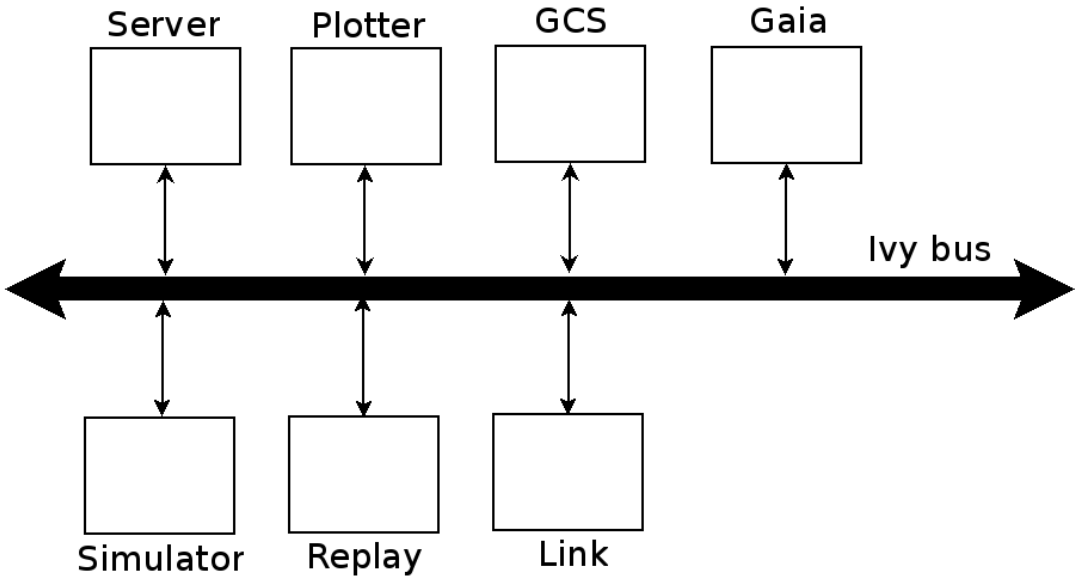
A Distributed System



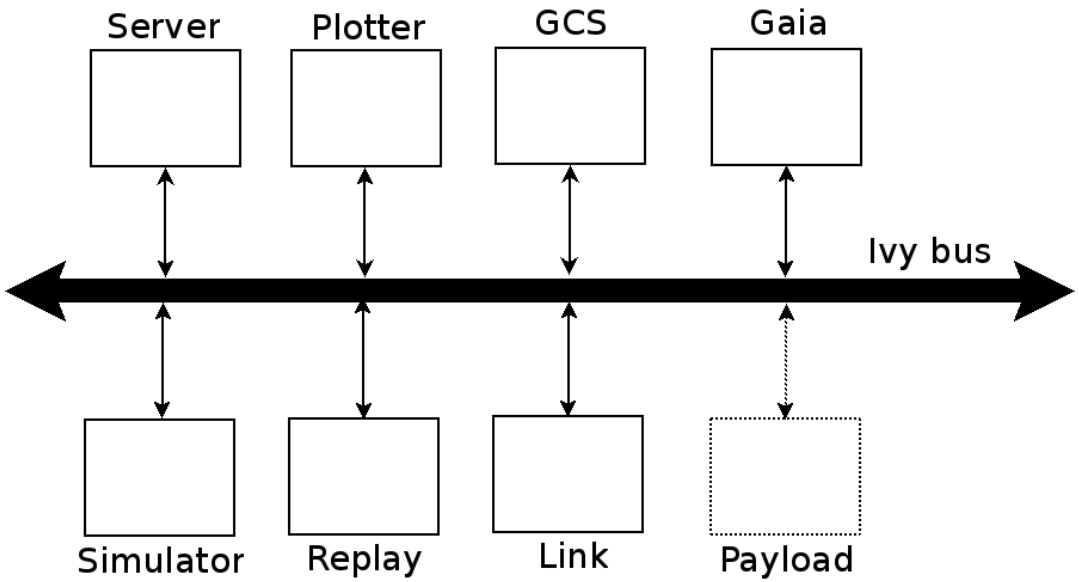
A Distributed System



A Distributed System

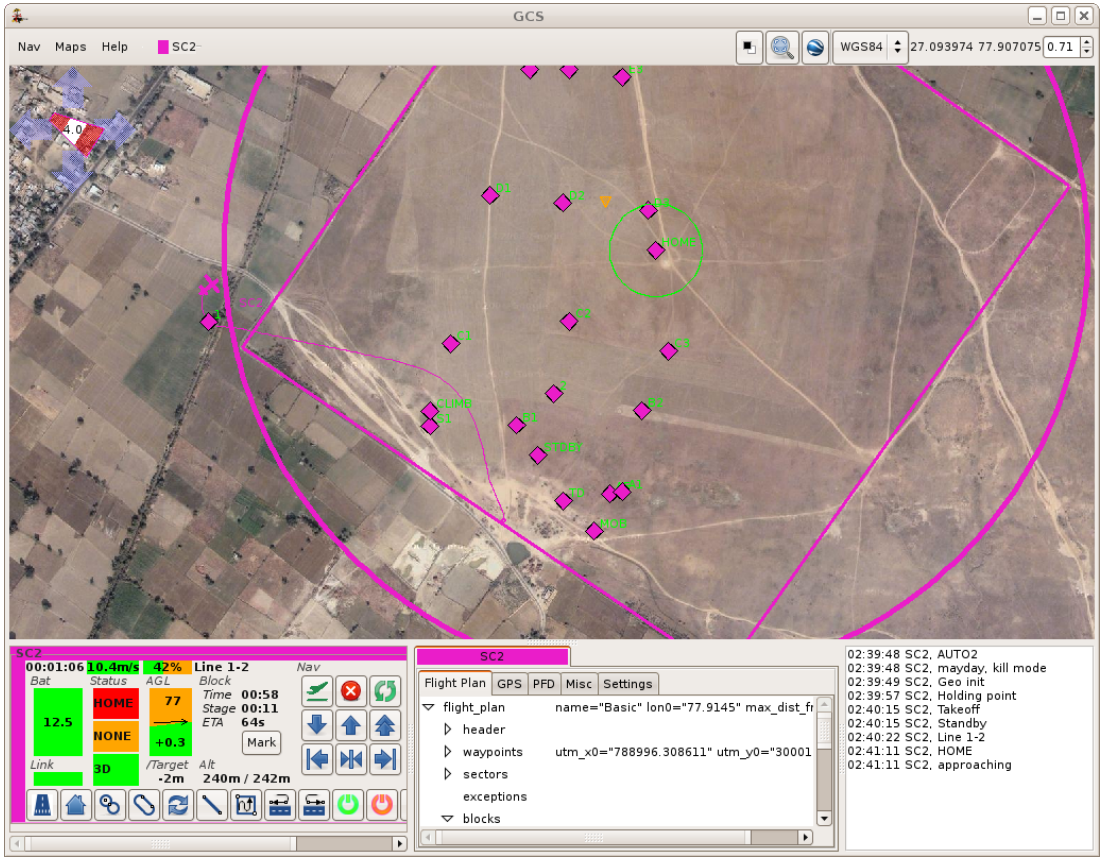


A Distributed System



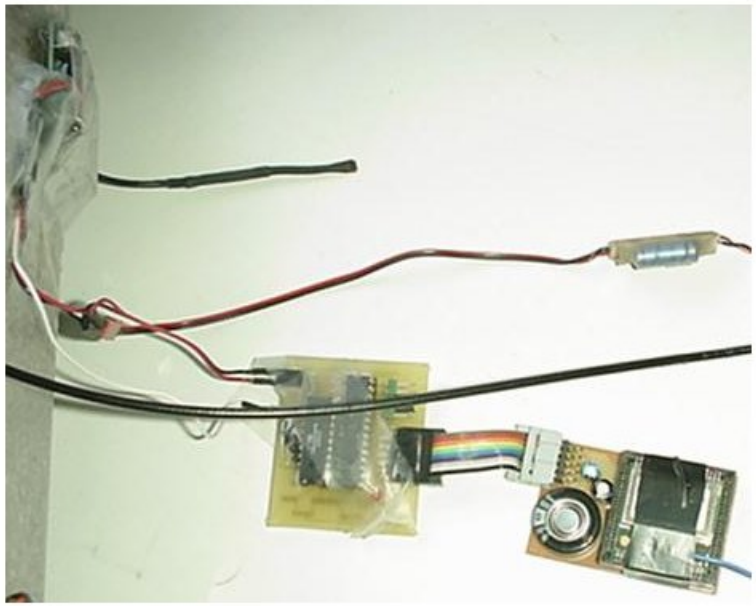
Safety

- Simulation
- Failsafe modes
  - Battery
  - NO GPS
  - Home
  - Kill
- GCS
- Extensive testing



History of the Project



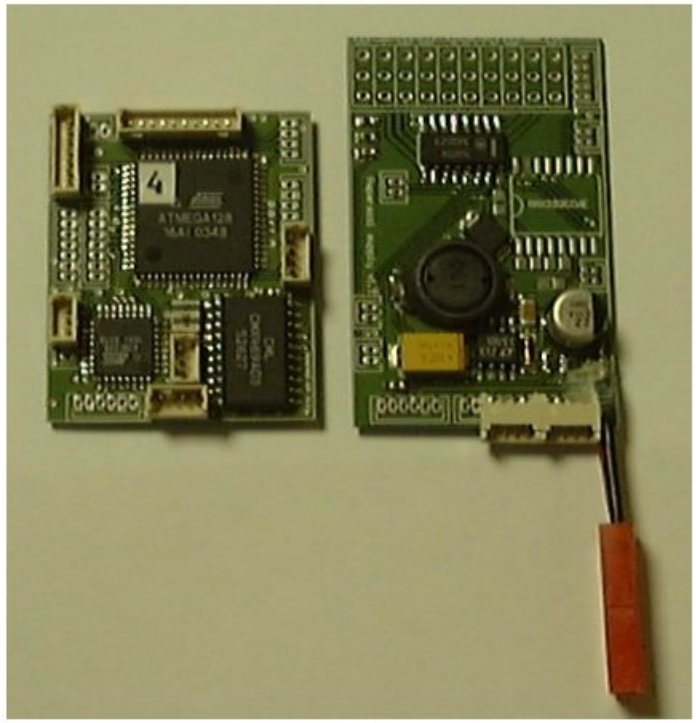


2003

History of the Project



2004



History of the Project



2005



History of the Project





2006

History of the Project





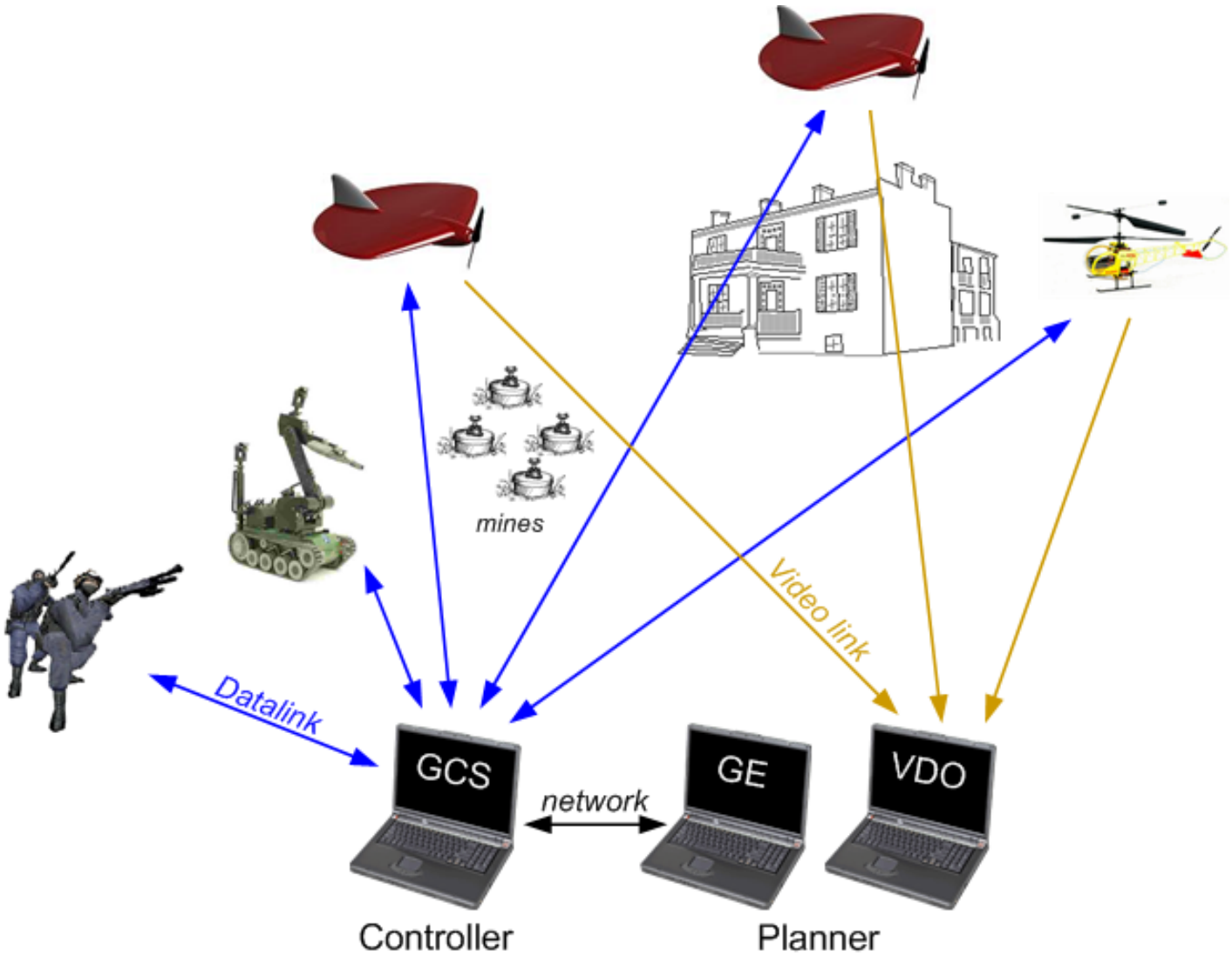
2007

Previous MAV Competitions

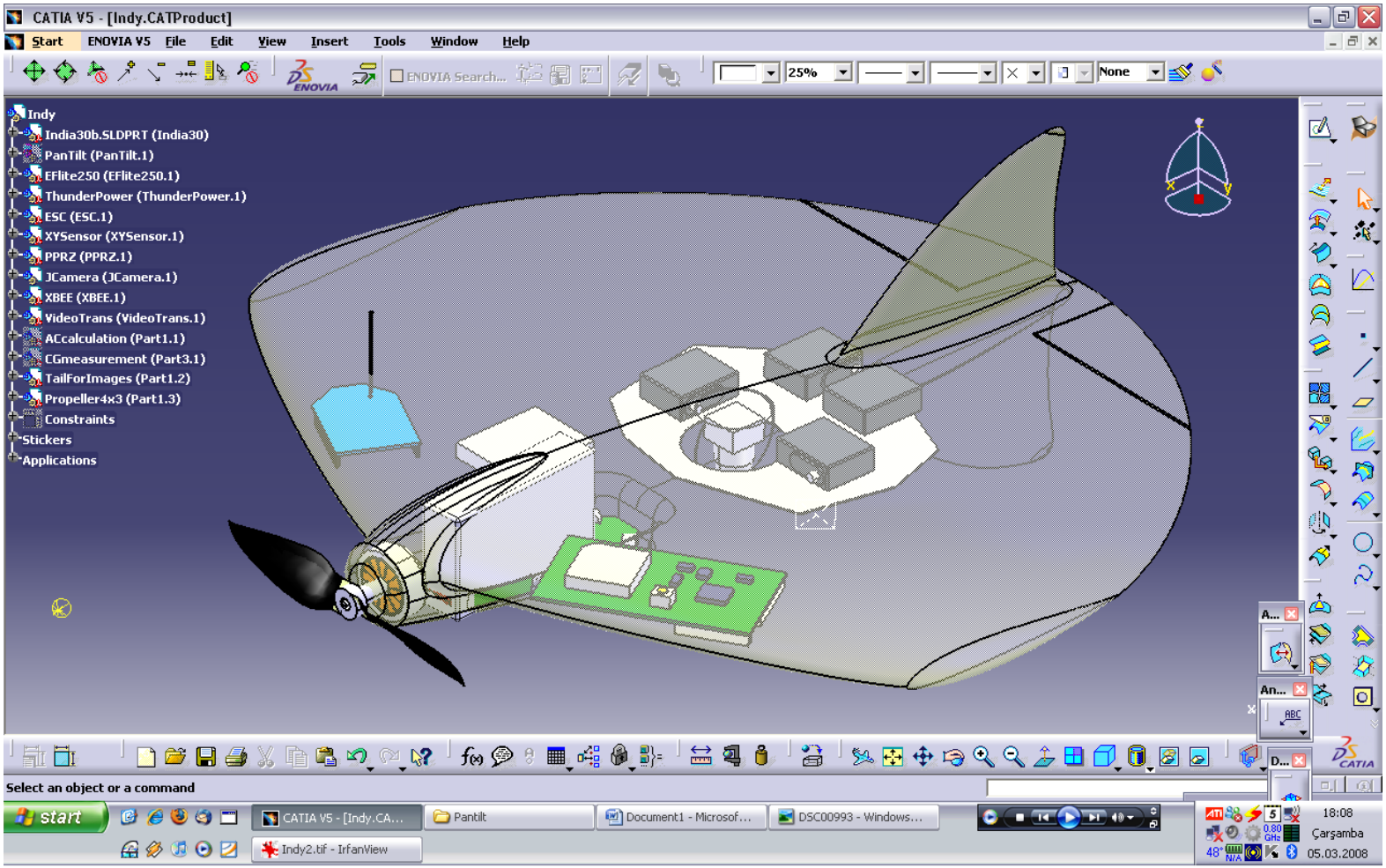


- JMD03, Toulouse, France: 1st place with the Twinstar
- EMAV04, Braunschweig, Germany: 1st place with the Microjet
- JMD04, Toulouse, France: 1st place with the Microjet
- MAV05, Garmisch, Germany: 4 Paparazzi teams at the first 4 places
- EMAV06, Braunschweig, Germany : all the teams were equiped with Paparazzi
- MAV06, Sandestin, Florida: 2nd and 3rd places
- MAV07, Toulouse, France: 1st place (tie), 3rd, 4th and 5th places

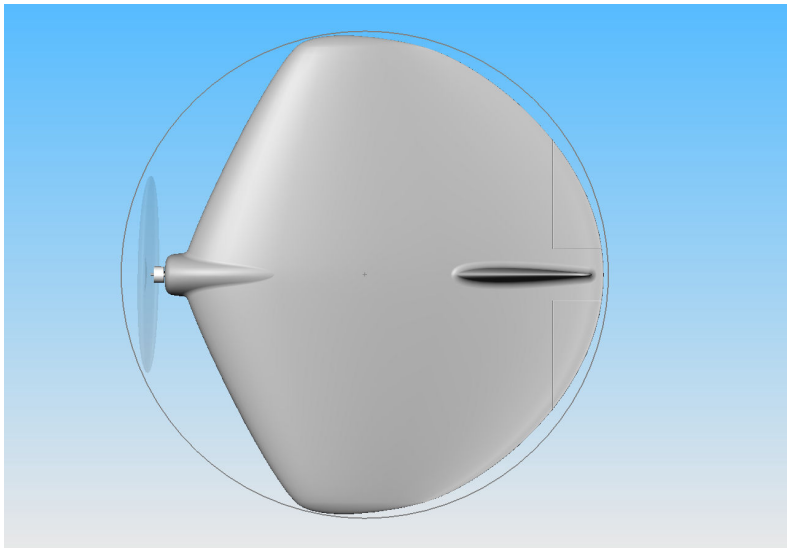
System MAV08

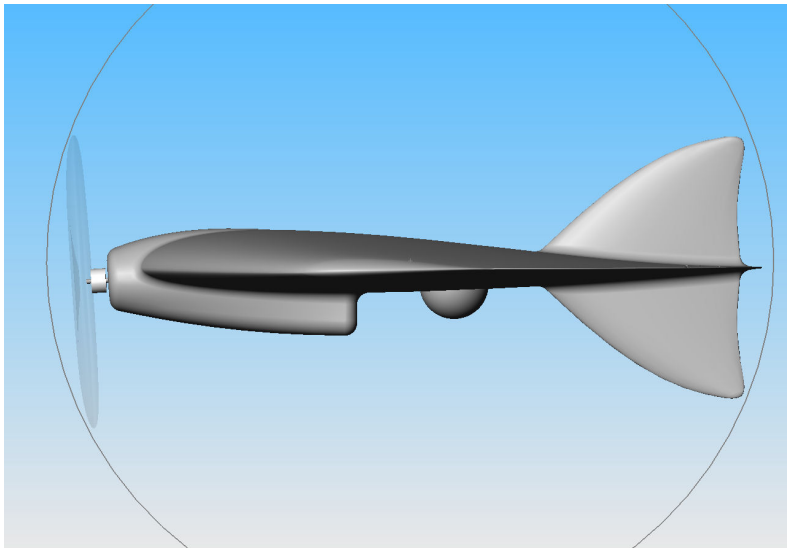


The Slicer

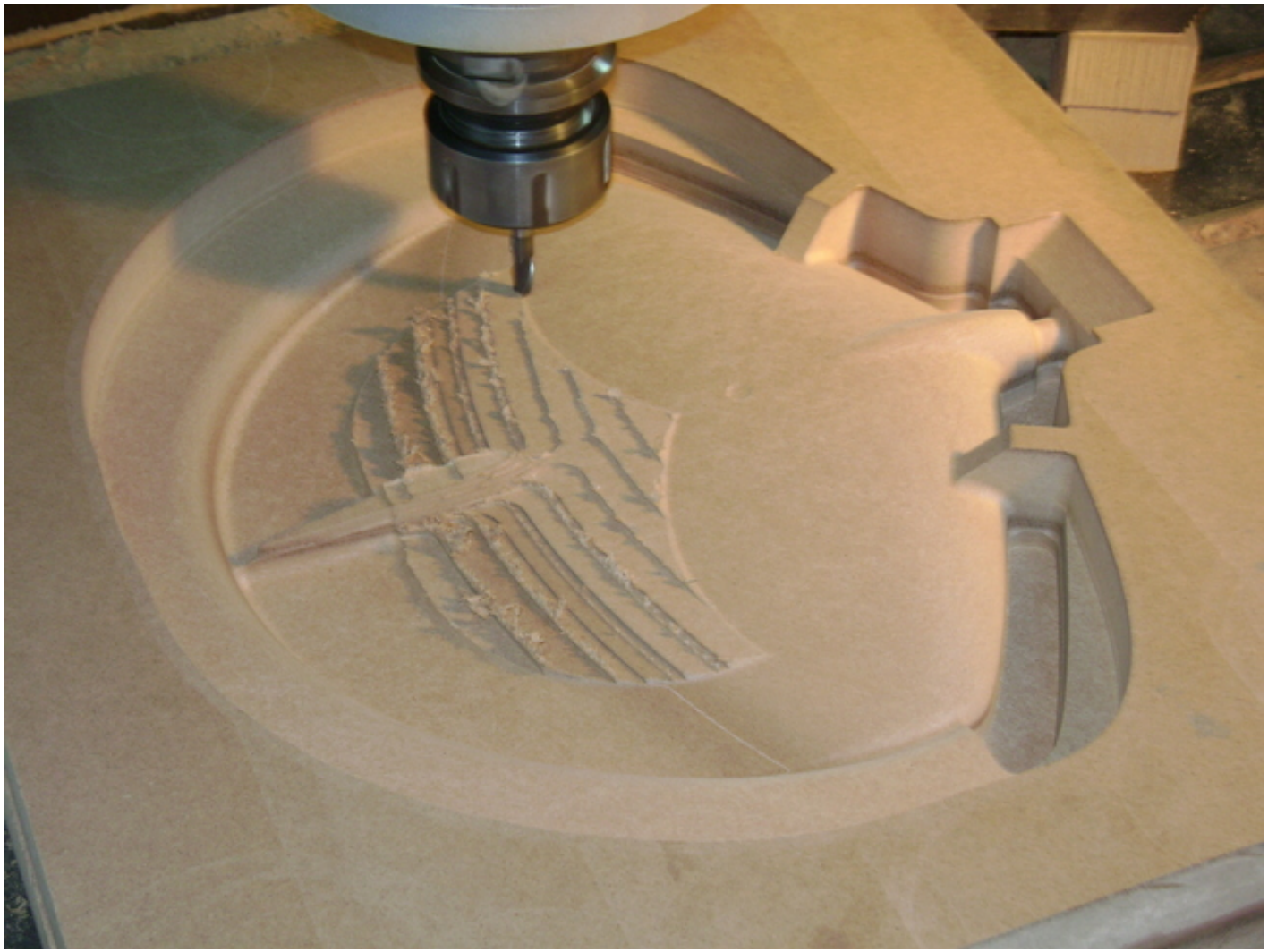

CATIA V5 - [Indy.CATProduct]
Start
ENOVIA V5
File
Edit
View
Insert
Tools
Window
Help
Indy
India30b.SLDPRT (India30)
PanTilt (PanTilt.1)
EFlite250 (EFlite250.1)
ThunderPower (ThunderPower.1)
ESC (ESC.1)
XYSensor (XYSensor.1)
PPRZ (PPRZ.1)
JCamera (JCamera.1)
XBEE (XBEE.1)
VideoTrans (VideoTrans.1)
ACcalculation (Part1.1)
CGmeasurement (Part3.1)
TailForImages (Part1.2)
Propeller4x3 (Part1.3)
Constraints
Stickers
Applications
Select an object or a command
start
CATIA V5 - [Indy.CA...
Pantilt
Document1 - Microsof...
DSC00993 - Windows...
Indy2.tif - IrfanView
18:08
Çarşamba
05.03.2008

The Slicer

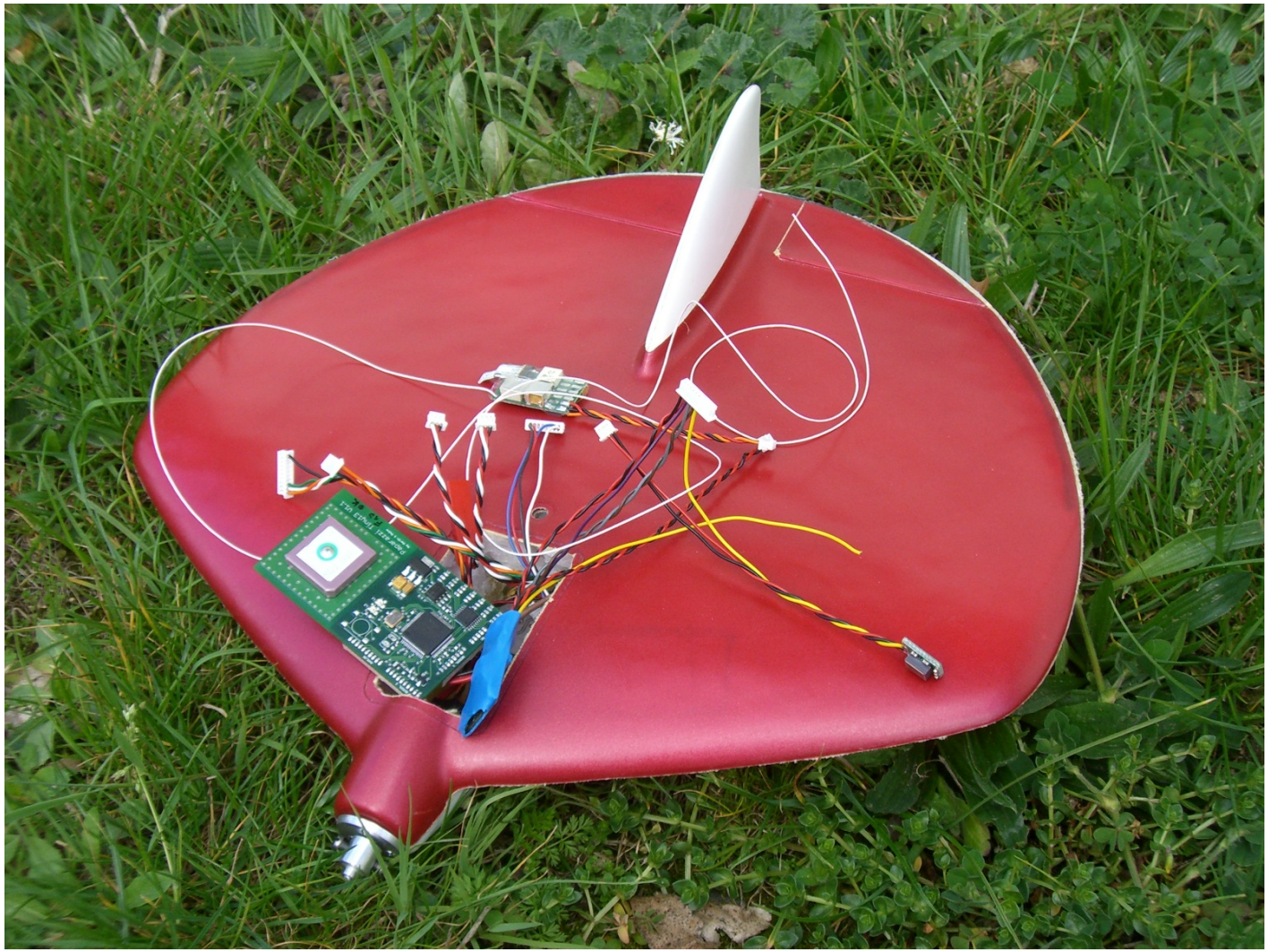




The Slicer



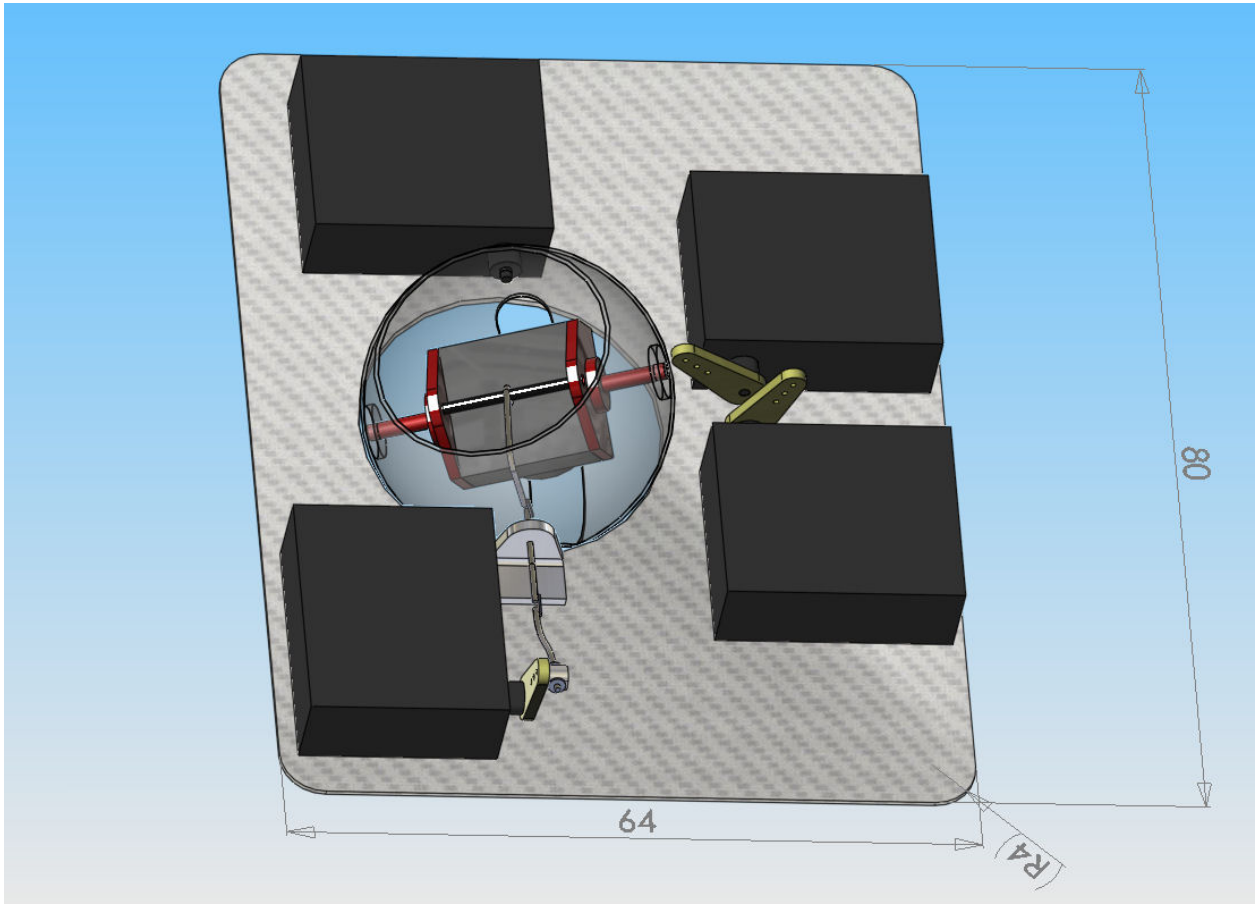
The Slicer



Weight Breakdown

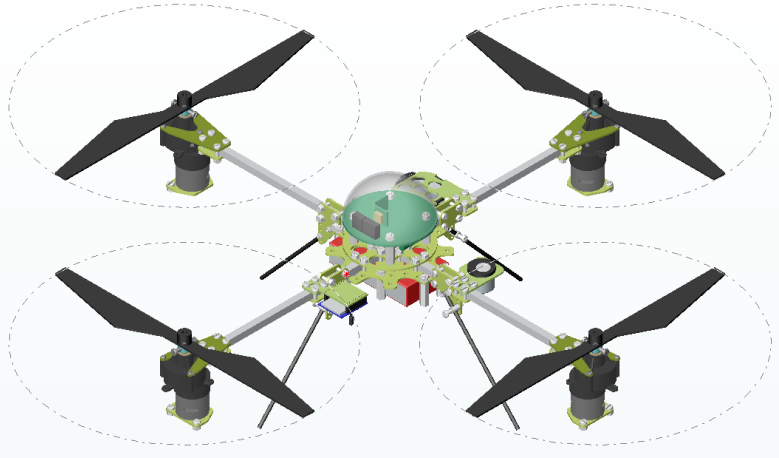
Part	Description	Grams
Airframe	Kevlar skins + EPP core	65
Propulsion	Brushless motor + Lipo battery + controller	85
Avionics	Autopilot + sensors + modems	45
Payload	Pan tilt camera + transmitter	30
Total		225

Mobile Camera



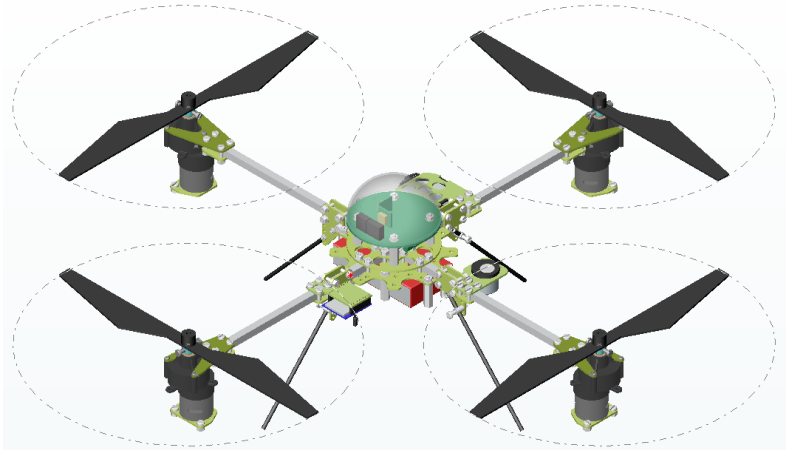
Helicopter

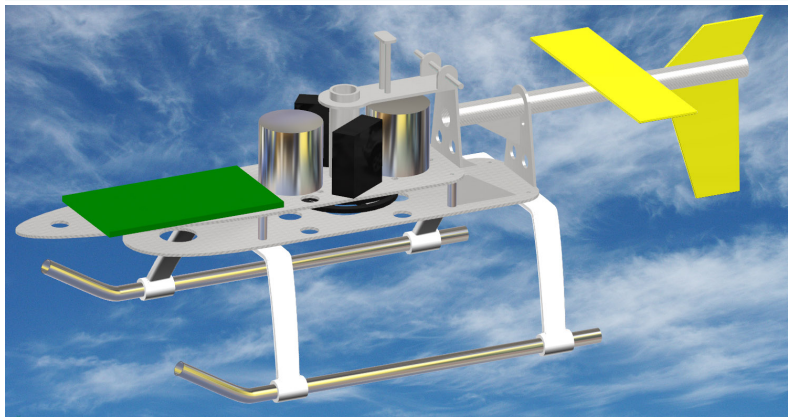
- Testbed



Helicopter

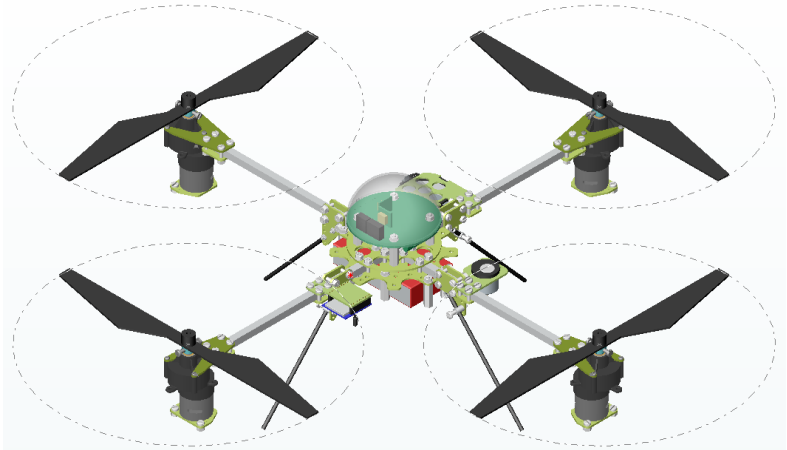
- Testbed
- 30cm Twisted Logic Coaxial Helicopter

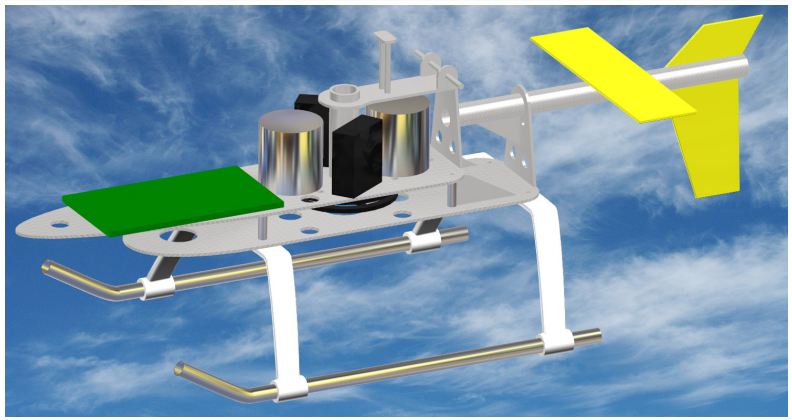




Helicopter

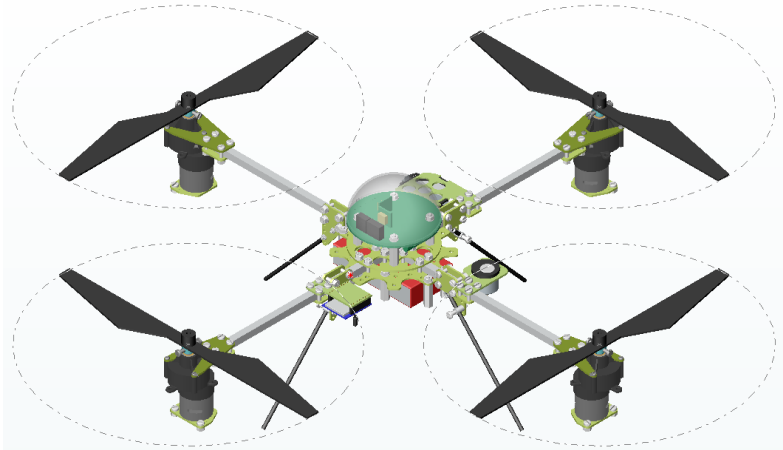
- Testbed
- 30cm Twisted Logic Coaxial Helicopter
- Control
  - Attitude from mechanics
  - Yaw stablization (gyro + magnetometer)
  - Vertical stabilization (accelerometer + barometer/rangemeter)
  - No navigation yet

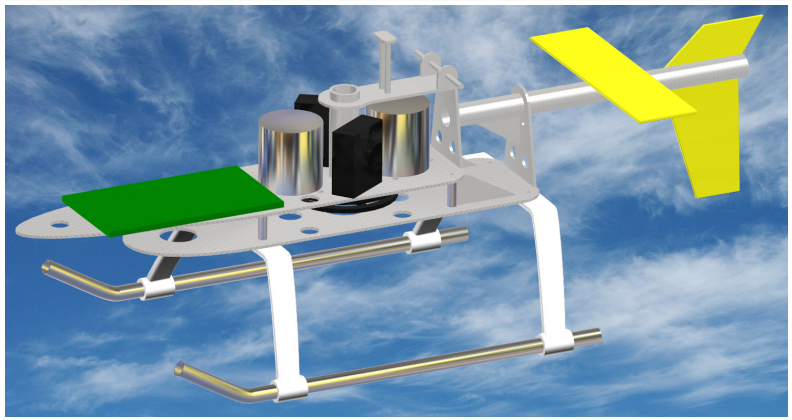




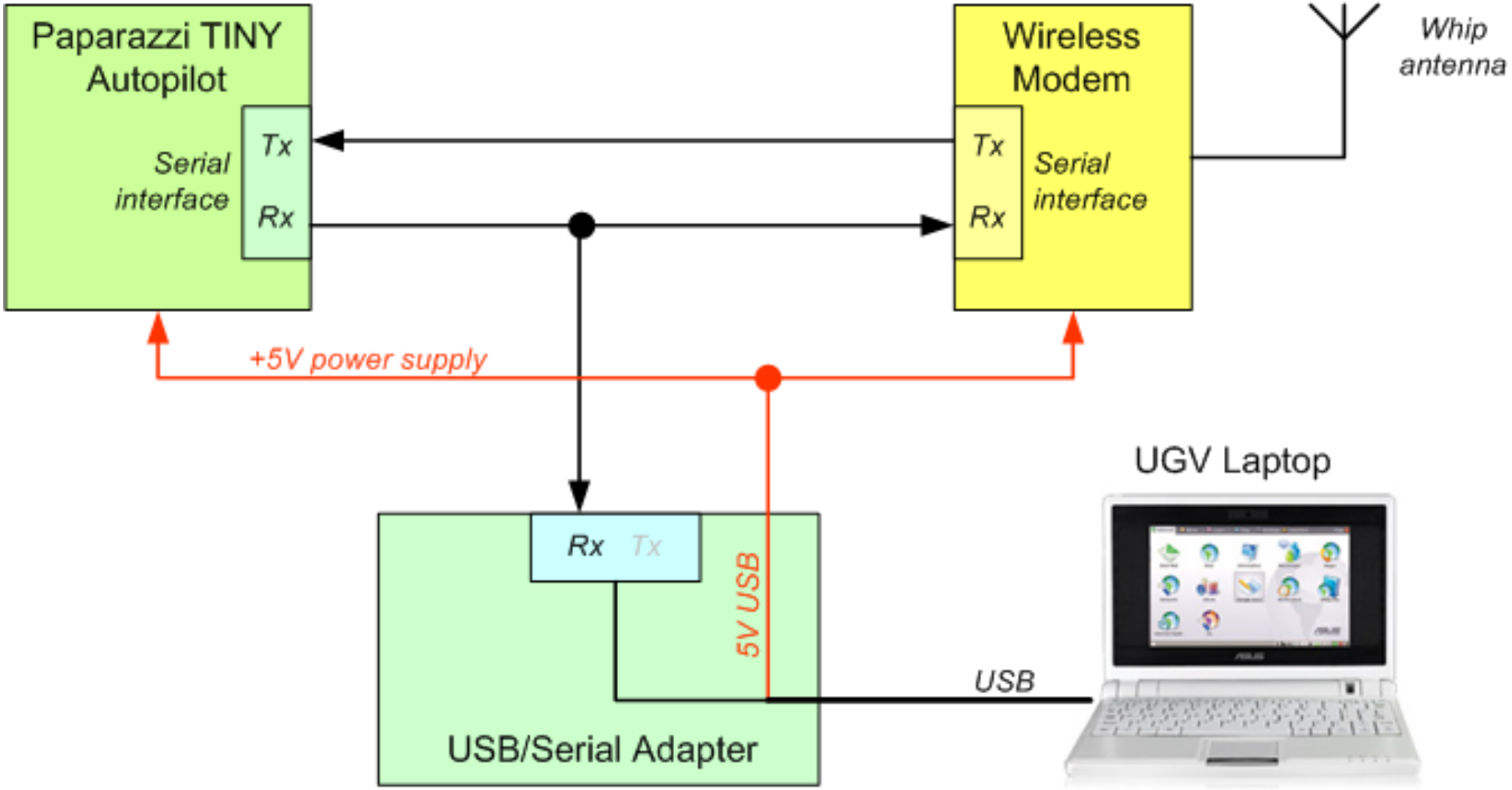
Helicopter

- Testbed
- 30cm Twisted Logic Coaxial Helicopter
- Control
  - Attitude from mechanics
  - Yaw stablization (gyro + magnetometer)
  - Vertical stabilization (accelerometer + barometer/rangemeter)
  - No navigation yet
- Not fast enough to safely fly outdoor

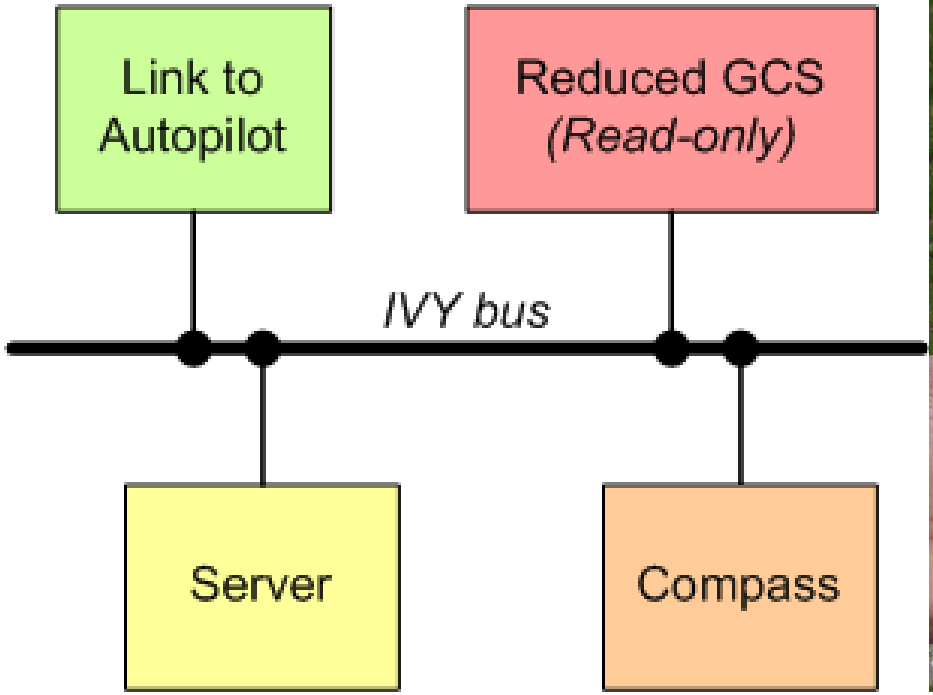




The UGV (Hardware)

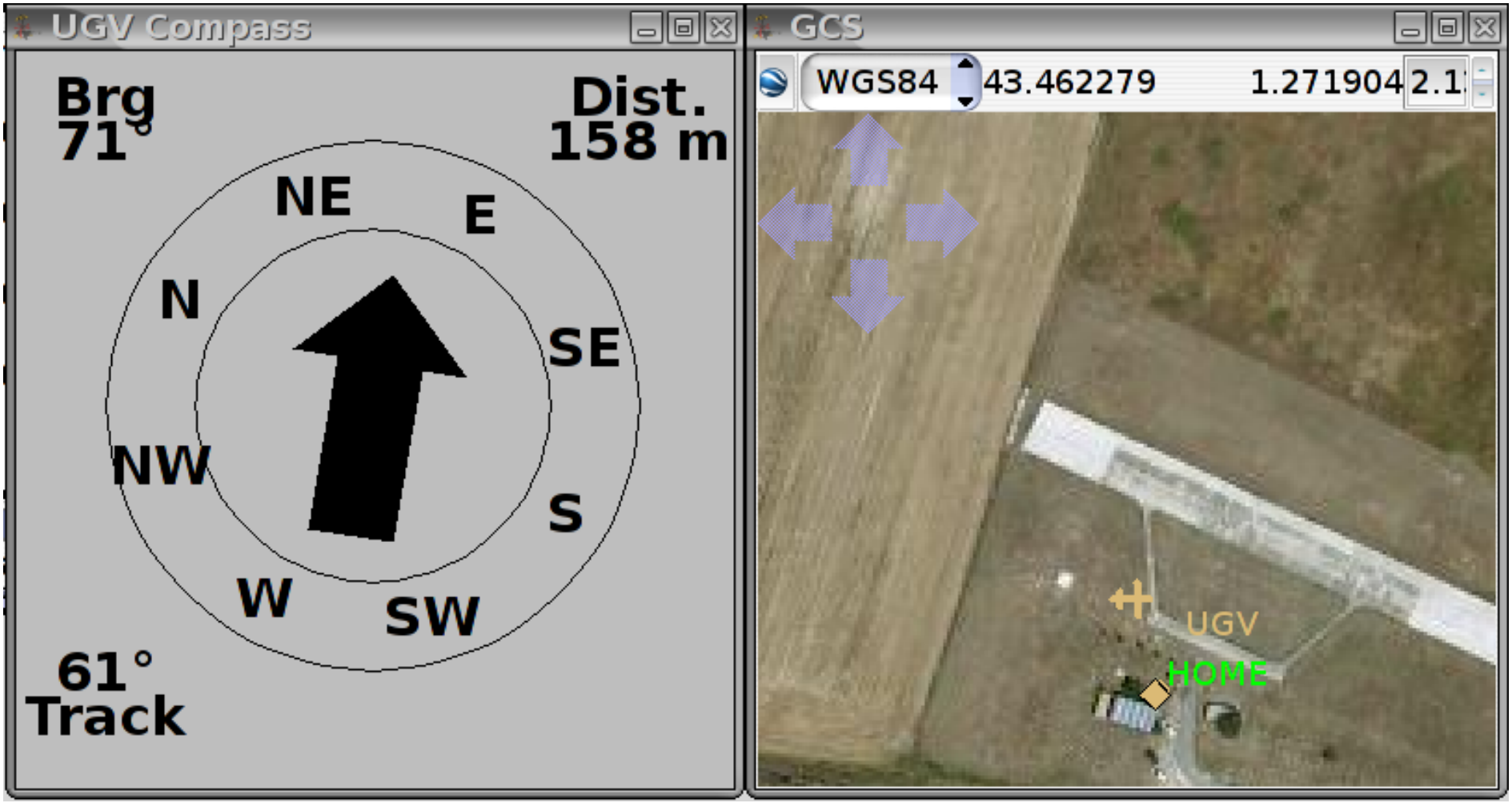


The UGV (Software)



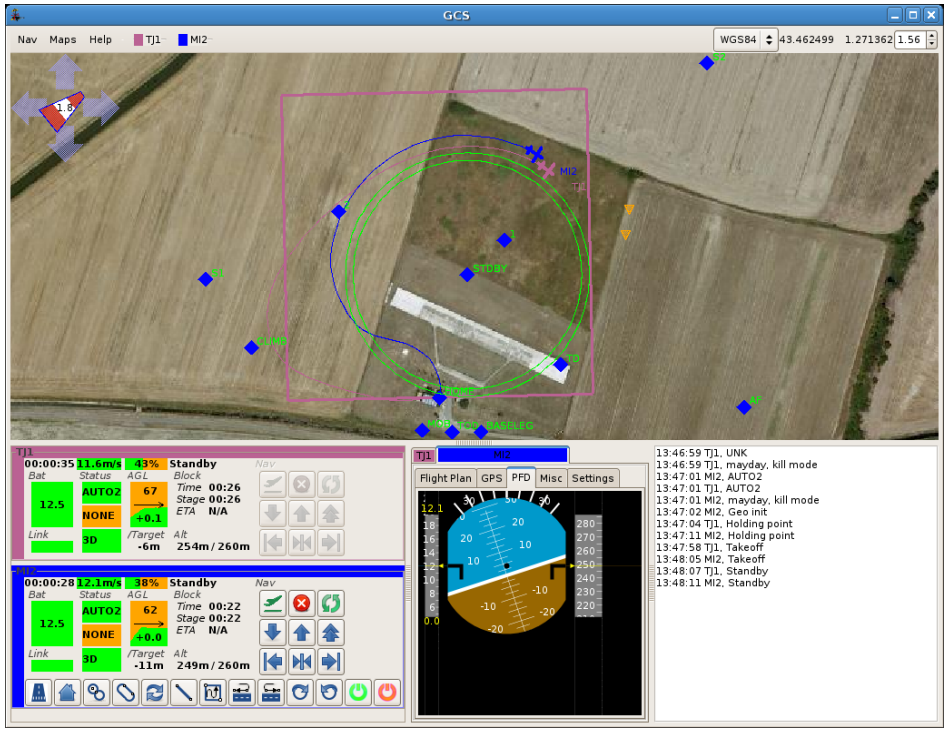


The UGV (Operator GUI)



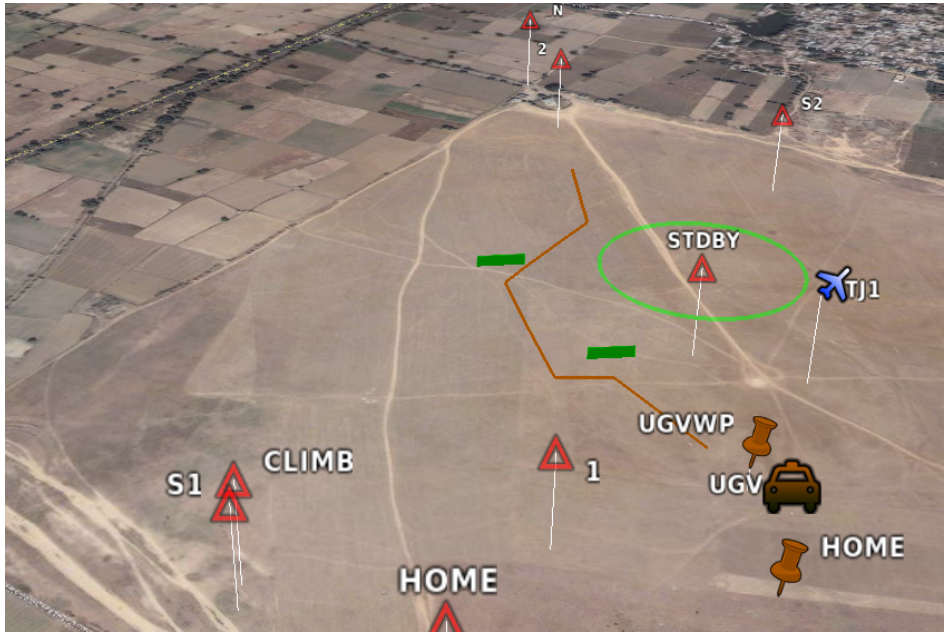
Operations

- GCS



Operations

- GCS
- Google Earth editor



Operations

- GCS
- Google Earth editor
- Video



Conclusion



- Two fixed wing vehicles

Conclusion



- Two fixed wing vehicles
- No autonomous rotorcraft

Conclusion



- Two fixed wing vehicles
- No autonomous rotorcraft
- Paparazzi equipped sim-UGV and Commando

Conclusion



- Two fixed wing vehicles
- No autonomous rotorcraft
- Paparazzi equipped sim-UGV and Commando
- One single GCS

Questions ?

Paparazzi: The Free Autopilot



paparazzi.enac.fr